Signal Processing and Feature Extraction in Markerless Telerehabilitation

Silvana G. Dellepiane, *Member, IEEE*, Federica Ferraro, Camilla Baffigo, and Marina Simonini

***Abstract*—Telerehabilitation solutions are a concrete answer to many needs in the healthcare framework since they enable remote support for patients and foster continuity of care. This paper explores telerehabilitation using the ReMoVES system, a markerless approach that facilitates remote exercise guidance. Focusing on the sit-to-stand (STS) task, which is crucial for daily activities, this study employs the Microsoft Kinect sensor for human movement monitoring. Emphasizing preprocessing and analysis, the research extracts reliable parameters, enabling remote observation and evaluation of patient performance. This study highlights the importance of noise reduction and automatic segmentation for feature extraction, which are essential for assessing task execution and identifying compensatory movements. By utilizing a diverse healthy subject group, a reference model is established, providing optimal features for accurate exercise execution. Statistical analyses involving both healthy subjects and patients revealed key features for remote exercise observation. Automatic feature extraction related to poses and body movements, together with homogeneity within control group sessions, forms the basis for a quantitative parametric model. This model describes and compares accurate exercise execution, offering a method to remotely evaluate and adapt individual rehabilitation plans on the basis of robust and reliable parameters.**

***Index Terms*—Telerehabilitation, markerless motion capture (MMC) sensor, skeletal joint detection, nonlinear signal processing.**

I. Introduction

IN a telerehabilitation context, the patient is provided with systems capable of remotely giving indications on the prescribed exercises while medical professionals can remotely observe the activity performed. An example of this approach is the markerless/contactless ReMoVES system [1], which is capable of acquiring a large amount of data and signals from which the skeletal joints of the patient's body are estimated and information of clinical interest is extracted. Based on Microsoft Kinect technology [2], ReMoVES offers specific exergames that encourage high repetition, provide guidance to the patient, and can be adapted to the needs of the individual rehabilitation plan. Through exergames, the automatic synchronization of the acquired signals is made possible while fully respecting the patient's time. The multidisciplinary rehabilitation team develops individualized prescriptions for patients, consisting of a detailed list and an exercise schedule. The system allows clinicians to monitor patient activities asynchronously. One major advantage of this user-friendly system is its ability to capture a high volume of data over time, particularly through repeated movements. This wealth of data allows for the extraction of significant and reliable features, supporting the evaluation of a patient's performance and progress. The large dataset is especially valuable for conditions requiring repetitive exercises, where movement accuracy and consistency are essential for successful rehabilitation. The present work aims to prove that a remote evaluation of patient activity is possible through a set of appropriate features capable of describing the correct execution model with good reliability and sufficient accuracy. The ability to safely rise from a chair is considered a fundamental prerequisite for activities of daily living (ADL) and may be severely impaired in poststroke individuals and frail older adults. Rehabilitation guidelines suggest repeated practice, mentally or physically, of sit-to-stand (STS) tasks [3]. Such rehabilitation interventions can be effective in improving STS performance, body symmetry, and the ability to stand independently. While much of the literature has focused on filtering RGB and depth videos from the Kinect, fewer studies have addressed the filtering of skeletal joint signals. Although noise affecting Kinect signals can arise from various sources, such as jitter, flickering, and missing samples, the existing literature primarily discusses only the smoothing of jitter effects. In this paper, we shift our focus to missing samples, alongside the issue of smoothing, providing a formal description that considers both additive and multiplicative noise. The proposed filtering method comprises a two-stage process involving interpolation and smoothing. Such preprocessing and analysis of the acquired signals are prerequisites for subsequent feature extraction. On the basis of the underlying signal model, after reducing the noise effects,

Received 22 April 2024; revised 5 November 2024 and 17 January 2025; accepted 3 February 2025. Date of publication 11 February 2025; date of current version 4 March 2025. This work was supported in part by the Piano Nazionale di Ripresa e Resilienza (PNRR) Project Robotics and Artificial Intelligence (AI) for Socioeconomic Empowerment (RAISE) funded by European Union's NextGeneration. *(Corresponding author: Silvana G. Dellepiane.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Scientific Board and the Regional Ethical Committee under Application 10109-n. 338.

Silvana G. Dellepiane is with the Dipartimento di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni (DITEN), Università degli Studi di Genova, 16126 Genoa, Italy, and also with the RAISE Ecosystem, 16122 Genoa, Italy (e-mail: silvana.dellepiane@unige.it).

Federica Ferraro and Camilla Baffigo are with DITEN, Università degli Studi di Genova, 16126 Genoa, Italy.

Marina Simonini is with the Struttura Complessa di Recupero e Rieducazione Funzionale (SCRRF), ASL3 Azienda Sociosanitaria Ligure, 16125 Genoa, Italy.

This article has supplementary downloadable material available at https://doi.org/10.1109/TNSRE.2025.3541153, provided by the authors.

Digital Object Identifier 10.1109/TNSRE.2025.3541153

automatic segmentation of the most relevant signal portions is proposed to arrive at the possibility of extracting robust and reliable parameters. A data engineering process is then proposed to assess the statistical significance of available features for evaluating the accurate execution of the required task, detecting any potential compensatory movements and assessing the overall functionality level. To construct a reference model and identify features that most effectively characterize exercise performance quality, a sample of healthy subjects with diverse characteristics (age, sex, physical condition, etc.) was utilized. Despite the sample's heterogeneity, a model describing correct execution performance was established on the basis of the optimal features exhibiting substantial homogeneity. Using this control group along with a group of frail elderly individuals and a group of poststroke patients, various statistical analyses of signals and key indicators were conducted to pinpoint the most relevant features for remote observation of exercise execution and discrimination. Several features referring to significant patient poses and movements have been considered, and their automatic extraction has been suggested through appropriate preprocessing, segmentation, and suitable similarity metrics. The reliability and significance of these features have been investigated by proposing strategies to reduce signal noise, minimize errors by leveraging signal redundancy, and examine homogeneity within the control group. However, we also assessed the relationships and correlations between features, tested for significant differences between the control and patient groups and quantitatively evaluated compensatory movements. The identified homogeneity in the control group sessions provides a foundation for establishing a quantitative parametric model to describe and compare the accurate execution of a given exercise. Finally, the robustness of the extracted features was evaluated through statistical comparisons and effect size calculations, which demonstrated their ability to capture relevant distinctions across groups. Clinical interpretability was also assessed, ensuring that the features aligned with expected physiological differences between populations, reinforcing their practical relevance in a clinical context.

II. State of the Art

Despite the ongoing debate over the definition of the term "exergame" [4], we use it here to describe video games designed to promote physical exercise and support rehabilitation practices. As already highlighted in the 2015 Canadian Stroke Best Practice Recommendations [5], virtual reality, including both immersive and nonimmersive technologies such as gaming devices, can serve as an adjunct to traditional rehabilitation therapies, offering additional opportunities for engagement, feedback, repetition, intensity, and task-oriented training. After the key aspects of the STS exercise are outlined, this section II provides a review of previous works in similar applications. Finally, the most recent technologies used in rehabilitation, along with the ReMoVES system, are briefly introduced in subsections II-B and II-C.

A. Sit-to-Stand

The sit-to-stand exercise stands out as a foundational element in rehabilitation, acknowledged for its efficacy in assessing lower limb strength and functioning as a targeted strength-reinforcing task [6]. Its significance spans fall risk evaluation [7], poststroke assessments, and insights into postural control. The STS exercise demonstrates adaptability across diverse patient categories, including individuals with hemiparesis, with attention to nuanced compensatory movements. Previous research has highlighted the clinical relevance of various parameters during STS, including the transition count [8], trunk and tibia flexion angles [9], and the detection of erroneous movements. The execution of the exercise by healthy individuals serves as a benchmark, revealing deviations in poststroke subjects and providing valuable diagnostic indicators [10]. Integration of technology facilitates the development of exergames tailored to guide and engage patients in correct execution of the sit-to-stand and stand-to-sit transitions. Two STS protocols exist: the 30-s STS protocol, which consists of standing and sitting as many times as possible in a period of 30 seconds, and the 5-time STS protocol, which requires five complete STS cycles to be performed in the shortest amount of time. Here, the 30-s STS protocol is considered [11], and sessions for a group of healthy subjects, frail elderly individuals, and poststroke individuals are described.

B. Rehabilitation: Enabling Technologies

Rehabilitation is undergoing a transformative shift, moving beyond traditional methods to embrace technology-driven solutions. Motion capture, performance assessment, and range of motion (ROM) measurements have greatly benefited from new technologies that enhance traditional therapy and assessments, offering advanced capabilities beyond conventional instruments. Compared to traditional supervision by rehabilitation physicians, motion capture systems provide distinct advantages, including precise joint angle sequences, motion visualization, remote rehabilitation capabilities, and access to large databases for data analysis. Motion capture systems are primarily categorized into optical-based and inertia-based sensors [12]. Inertial measurement units (IMUs) consist of a gyroscope, an accelerometer, and a magnetometer, which together allow the determination of the orientation of the rigid body to which the sensor is attached. Thanks to the low-cost, portability, energy-efficiency, and wireless transmission, they are ideal for use as wearable devices in rehabilitation applications. Section I of the Supplementary File presents a concise overview of these latest technologies, with a particular focus on the most promising solutions for future home rehabilitation support, particularly inertia-based sensors and optical systems. According to [13], from among optical-based systems, the markerless motion capture (MMC) technology has particularly gained prominence in rehabilitation because of its cost-effectiveness, efficiency, and patient-friendly nature [14]. The portability and low-cost attributes of these systems contribute to obtaining kinematic data during sit-to-stand transitions, offering a basis for precise and quantitative assessments of functional parameters even in home-based settings. To provide a comparison with other MMC technologies, we start from the systematic review reported in [15], which examined 65 studies on the use of such a technology for clinical measurement in rehabilitation. A detailed description is provided in the Supplementary Files, Section I-B, focusing on the research's purpose of detecting differences in movement patterns between well-executed and poorly executed exercises. The extensive and effective use of Kinect within our area of interest is supported by the aforementioned review as well as a broader examination of the research literature.

C. ReMoVES

As anticipated at the beginning of this section, virtual reality adds significant value in the rehabilitation context, where specific exercises can be delivered to the patient without the presence of a physiotherapist. Video game consoles can simultaneously provide movement instructions, create an interactive environment, and monitor patient movements. Therefore, they are the preferred technology, even compared with video cameras and smartphones. One of the first works based on the Microsoft Kinect device dates back to 2011 [16] for balance training. Compared with other similar technologies (e.g., Nintendo Wii [17]), Microsoft Kinect is also a contactless system, as it does not require the use of remote controllers. Rehabilitation studies using Kinect have focused on poststroke and recovery [18], Parkinson's disease assessment [19], and orthopedic rehabilitation [20]. Kinect-based exergames have significant applications in assistive technologies for elderly individuals, including fall risk reduction [21], improving physical performance, and reversing the deterioration process in frail and prefrail elderly people.

In the current paper, the delivery of exercises and indications to the patient are provided by the ReMoVES IoT system (numip.it/removes), which is largely described in [1]. Designed to integrate with traditional rehabilitation, ReMoVES operates during periods without therapist-guided activity, enabling information collection and access from various locations. The ReMoVES IoT system features a server-client architecture comprising four traditional system layers: the sensor, network, server, and application layers. These layers are detailed in the Supplementary File, Section II. The Equilibrium Paint exergame, delivered by ReMoVES, considers the 30-s STS activity described in Section II-A, providing both assessment and muscular strengthening benefits. Since ReMoVES is based on the Kinect_v2 video game console and sensor, which acquires the patient's activities without the need for markers or invasive video cameras, its accuracy, ease of use, and cost-effectiveness are closely related to the considerations discussed previously. Only the depth data that allow the spatial tracking of the human body and the identification of 25 joints via Microsoft proprietary software are exploited. By developing game-based rehabilitation tools that are tailored to the therapy goals of different patient categories, it is possible to provide clinicians with meaningful performance data [22]. With respect to the evaluation of the Kinect depth sensor accuracy, several studies have described the error as a function of the distance of one object from the camera. Ref. [23] reported a comparison between the former generation, Kinect_v1, Kinect_v2, and the more recent Kinect-Azure, showing depth errors on the order of a few millimeters. With reference to the error in estimating the position of the joints, the most exhaustive study was reported in [24], where the authors reported the measured difference between Kinect_v2 and the video-based Qualisys motion capture system. Other studies [25], which compared the Kinect system with the Vicon system, confirmed that the measurement accuracy was better in the mediolateral axis than in the vertical and frontal axes. Some joints (i.e., ankles and hips) have larger errors than others. Although the measurements obtained in the experimental sessions reported in the present work seem to suggest a lower error, as discussed in subsection VI-B, some values from [24] will be reported with the aim of describing the improvements obtained by the present approach.

III. Human Pose and Movement Analysis

The signals that are able to describe patient movements are introduced hereinafter, along with a brief description of noise that usually affects such acquisitions. The main features that can be extracted from the acquired signals are then defined. The Kinect_v2 sensor utilizes time-of-flight (ToF) technology, recognized as the standard for understanding human motion. A detailed description of the Kinect_v2 sensor, with particular emphasis on its human capture function is given in Supplementary Files, Section I-C.

A. Kinect Joint Signals

A stochastic (or random) process is a correspondence that associates a temporal function with each possible result of an experiment. Each repetition of the experiment (i.e., a trial) gives rise to one of the process realizations. The 3D signal referring to the $j-th$ joint acquired in space (x, y, z) by the Kinect's infrared depth sensor is as follows:

$$\{\mathbf{v}_j(t)\} = \begin{bmatrix} \{x_j(t)\} \\ \{y_j(t)\} \\ \{z_j(t)\} \end{bmatrix} \tag{1}$$

with $j \in \{1, \ldots, 25\}$. More precisely, this is a digital signal $\{\mathbf{v}_j(t)\} = \{\mathbf{v}_j(kT_s)\}$, with sampling frequency $f_s = 30[Hz]$ and $T_s = 0.0\overline{3}[s]$. Starting from these basic joint signals, other useful signals can be obtained for kinematic analysis of a person's movements. For example, displacements and trajectories in 3D space, on a plane, or along an axis, as well as speed, acceleration, etc., can be instantly measured. The distance, speed and acceleration of each point or set of points can be studied in the same way. From the 3D coordinates, any angle can also be measured. A temporal feature signal $\psi(t)$ can then be measured and studied, which can be one of the basic signals or any derived measures. Assuming that the subject is in a given postural state Γ, some characteristics of interest can be observed. Accordingly, we define a resting state, R_β, and a specific motion, M_α, to indicate specific resting positions or certain motions during which we can extract features. In the former case, one can refer to *Instant Features*, i.e., $\psi(t)|R_\beta$, and in the latter case, to *Dynamic*

Features, i.e., $\psi(t)|M_\alpha$. Given a random process, the analysis of the statistical properties of any order is meaningful only if the stationarity property is verified. In a complex pattern like the one depicted in a signal representing human movement, stationarity holds only if the signal is conditioned to a specific rest or dynamic state. For the subsequent definition of signals and noise models, it is therefore necessary to define some working conditions, which are also fundamental in the subsequent signal processing and analysis phases. The three components of the signal $\{\mathbf{v}_j(t)\}$ are independent, and their mean power, under the stationarity hypothesis, is S^2_{ij}. For instance, along the i axis, the mean power of joint j conditioned to a generic state Γ is $S^2_{ij}|\Gamma = E\{v^2_{ij}(t)|\Gamma\}$. Stationarity is also a precondition for verifying the ergodicity property that allows the use of a temporal mean operator in place of statistically expected values. Through an analytical reconstruction process, markerless sensors are able to produce 3D signals for a series of points of the human body that do not necessarily belong to the external surface but refer to a skeletal representation. The mediolateral (ML) axis x is positive to the right of the user; y is the vertical (VT) axis, positive upward; and z is the anterior–posterior (AP) axis, positive from front to back of the user. The major signal noise sources in real applications are classical additive noise and multiplicative noise, as described in Equation 2 for the $j-th$ joint acquired by a generic sensor as an example:

$$\{\mathbf{v}_j(t)\} = \{v_{ij}(t)\} = \{(\mathbf{v}_{Bj}(t) + \mathbf{n}_j(t))\} \cdot \{\mathbf{m}_j(t)\} \qquad (2)$$

where $\mathbf{v}_j(t)$ is the acquired signal, $\mathbf{v}_{Bj}(t)$ is the noise-free signal, $\mathbf{n}_j(t))$ is the additive noise and $\mathbf{m}_j(t)$ is the multiplicative noise used to describe jitter and nonlinearities. Noise affecting the Kinect signals is described in Section IV-C.

B. Kinect Noise Filtering

Kinect_v2 uses a machine learning approach, specifically a random forest classifier trained on a large dataset of human poses, to estimate 3D skeletal points from depth camera data. Common noise effects that challenge skeletal tracking include jitter, flickering, and missing samples. Jitter causes small, rapid oscillations in joint positions due to sensor noise or classification errors, resulting in shaky and unstable motion. Flickering involves noticeable jumps in joint positions, often caused by occlusion or inconsistencies in the classifier, leading to erratic motion. Missing samples occur when joint data are lost for certain frames, causing gaps in the skeletal data, making tracking incomplete or discontinuous, with joints disappearing and reappearing intermittently. The primary focus of studies published on filtering Kinect signals is to reduce jitter and smooth skeletal movement data. In [26], moving average, high-pass, and Kalman filters were evaluated for enhancing movement visualizations while ensuring their validity for physiotherapeutic assessments. Reference [27] presented a general overview of approaches for optimizing motion capture systems other than Kinect and discussed filtering methods. Excluding deep learning, which requires large training databases, in addition to the above filters, Savitzky–Golay and one-euro filters are used. In general, the Kalman filter, which is commonly used for marker-based and IMU-based systems, is especially useful for real-time filtering. The Savitzky–Golay filter applies local polynomial fitting, whereas the one-euro filter adjusts its cutoff frequency on the basis of signal velocity. These filters are simpler and less resource intensive than deep learning methods but require expert tuning, and their performance can vary. Indeed, in the absence of training sets with ground-truth gold standards, the evaluation relies on visual observation and the minimization of basic intrinsic parameters such as the mean absolute error, lacking external validation. This highlights that optimizing motion filtering remains a key research challenge. According to Winter [28], angular kinematic signals are generally characterized as low-frequency signals contaminated with high-frequency noise, which must be filtered using a 2nd- or 4th-order Butterworth low-pass filter. The ideal cutoff frequency should be determined through residual analysis, with the relevant spectral range for gait signals being below 6 Hz, whereas for running, it is below 25 Hz. The motor tasks of lateral trunk inclination, trunk rotation, and shoulder abduction, performed during heading, skiing, and goalkeeper Kinect games, were analyzed in [29]. Owing to a residual analysis based on Fourier analysis and empirical mode decomposition (EMD), the artifacts contaminating the kinematic signals collected with the Kinect have spectral characteristics of white noise. The ideal cutoff frequencies are between 3 and 5 Hz, which are lower than those used in previous studies, i.e., between 6 and 8 Hz [24], [30], [31].

C. The Features

The ReMoVES system captures data and signals from each game session, including meta-features like the game name, start and end times, final score, and additional game parameters to support efficient database querying. Beyond patient prescriptions and other meta-features not addressed in this study [32], the primary features analyzed here include kinematic instant features and dynamic features. An automatic signal segmentation step is proposed, as described in the next subsection, which automatically detects the start and end times of each repetition of the basic STS movement. After this automatic localization of the instantaneous time referring to some specific positions of the subject, the precise measurement of each joint constitutes the instant features that allow one to understand the posture and compare it with the posture indicated by the medical staff through the exergame. Obviously, in the STS, the two main instant features are related to standing and sitting positions. In particular, angles extracted from joint positions are of clinical interest. Dynamic features describe the kinematic movement that the subjects performed and refer to the ROM in a given period of interest, the actual temporal displacement of the joints, or the change in angles during a specific movement. The base features are the joint positions at an instant point, which are directly accessible by the signal punctual value. Even though coordinates in 3D space are available for each joint at a given time, their position on only one axis or anatomical plane is often of interest. For instance, in the current case, the vertical axis is of major importance. Starting from base features, the most relevant derived feature extracted by processing and combining the acquired signals is the distance between joints and instantaneous angles. As an example, the difference between the j and k joint positions on the i axis is defined as $\Delta i(t) = v_{ij}(t) - v_{ik}(t)$. Angles are computed by applying the absolute or relative angle formula, depending on the specific analysis. An absolute angle of a body segment linking two joints j and k in a given body plane is measured with respect to one of the axes by using the arctangent formula. As an example, the trunk flexion angle (in the sagittal plane) is defined here as the absolute angle of the trunk segment with respect to the antero–posterior axis in the body frontal direction and can then be measured as:

$$\theta_{FA-trunk} = arctan\left(\frac{\Delta y}{-\Delta z}\right) = arctan\left(\frac{y_{SPS} - y_{SPB}}{z_{SPB} - z_{SPS}}\right) \quad (3)$$

where SPS and SPB are the spine–shoulder and spine–base joints, respectively. The negative sign takes into account that the z axis points in the backward AP direction instead of the frontal AP direction. After the angle error is analyzed in detail, a better way of measuring the angle is suggested to reduce the measurement error. The internal angle at the knee is calculated as a relative angle because it corresponds to an articulation between three joints. The formula used is as follows:

$$\theta_{knee} = arctan\left(\frac{\|\mathbf{h} \times \mathbf{a}\|}{\mathbf{h} \cdot \mathbf{a}}\right) \quad (4)$$

where $\mathbf{h}$ and $\mathbf{a}$ are the knee–hip and knee–ankle vectors, respectively. Starting from the portions of signals related to a given movement, additional dynamic features are extracted. For example, the speed of a joint in space, its trajectory, the ROM of displacements or angles, and the instantaneous variations of a feature during a movement can be derived from the base signals.

1) Feature Standard Error: For each joint signal, the root-mean-square error (RMSE) is the instantaneous measurement error of the ideal signal, i.e., $v_{Bij}(t)$, compared with the acquired signal $v_{ij}(t)$ and is defined as:

$$\epsilon_{v_{ij}} = \sqrt{\epsilon^2_{v_{ij}}} = \sqrt{E\{(v_{ij}(t) - v_{Bij}(t))^2\}}, \quad (5)$$

ϵ_{ij} for the sake of brevity.

The error of the derived features follows the well-known error propagation rule. For the difference Δi, the mean square error is $\epsilon^2_{\Delta i} = \epsilon^2_{v_{ij}} + \epsilon^2_{v_{ik}}$. By recalling the absolute angle formula in Equation 3, the angle MSE is then:

$$\epsilon^2_\theta = \frac{\epsilon^2_{\Delta i}}{\Delta l^2} + \frac{\epsilon^2_{\Delta l} \cdot \Delta i^2}{\Delta l^4} = \frac{\epsilon^2_{\Delta i} + \epsilon^2_{\Delta l} \cdot tan^2(\theta)}{\Delta l^2} \quad (6)$$

where $\Delta l = L{\cdot}\cos(\theta)$, given that L is the segment length. As a consequence, the angle standard error depends on the error of the measured joints difference and on the magnitude of Δl. In particular, the error becomes large when Δl is small, which is also the case with angles close to $+/- \pi/2$. In addition, as one can expect, a smaller error is found when dealing with long body segments than when dealing with short ones. In conclusion, for angles in the interval $\pi/4 < |\theta| < 3\pi/4$, the error quickly diverges, and the arccotangent formula is proposed here instead of the arctangent formula in these cases to preserve small errors. As a consequence, the following approach is applied:

$$\theta\left(\frac{\Delta i}{\Delta l}\right) = \begin{cases} arctan\left(\frac{\Delta i}{\Delta l}\right) & |\Delta l| > |\Delta i| \\ arcotan\left(\frac{\Delta l}{\Delta i}\right) & |\Delta i| > |\Delta l| \end{cases} \quad (7)$$

With respect to error evaluation, when a feature is measured, the possibility of having independent repeated measurements allows for a better estimation. In descriptive statistics, the sample mean is the optimal estimate of the central tendency in such cases. The sample mean of N repetitions reduces the square error by a factor equal to N. The standard error is thus reduced by a factor equal to $\sqrt{N}$.

IV. Materials and Methods

A. Dataset

The present work is a retrospective study conducted in collaboration with the Local Health Authority ASL3, Sistema Sanitario Regione Liguria, Italy, specifically with Struttura Complessa di Recupero e Rieducazione Funzionale (SCRRF), directed by Marina Simonini. The study involved patients from La Colletta Hospital in Genoa, healthy subjects from the DITEN department and the same hospital, elderly residents in nursing homes affiliated with ASL3, and patients who underwent telerehabilitation at home. This research study was conducted following the principles of the Declaration of Helsinki and the approval of the Institutional Scientific Board and the Regional Ethical Committee (study ID 10109 - n. 338, November 25th, 2019). Written informed consent was obtained from the participants before the study began. This work presents a feasibility study to evaluate the possible use of ReMoVES for STS exercises in various real-world scenarios, including rehabilitation centers, retirement homes, and patients' homes. A control group is used to study the meaningfulness and reliability of the features of interest, which will later serve as a reference when working with poststroke and elderly subjects. The control group consisted of sixteen healthy subjects. The participants included 9 females and 7 males aged between 27 and 61 years, with an average age of 38.62 ± 12.95 years. Some of them performed the STS exergame several times, for a total of 26 sessions. Two poststroke patients (i.e., Group A) learned to use the system while hospitalized at La Colletta Hospital and then continued using it for an extended period at home. Thanks to its user friendliness, in both cases, no technical intervention was necessary for the installation, which was carried out by the patient and his family. During that time, a total of 65 sessions were recorded for the STS activity. The remote observation allowed the therapist to evaluate the activity carried out and suggest specific indications for a more correct execution. One patient was a 62-year-old man who performed the prescribed exercises for upper and lower limbs for a period of 39 days. A total of 24 sessions of STS activity were executed. This patient was evaluated in previous studies for his performance during upper limb rehabilitation exercises, as reported in [33]. The second patient was a 79-year-old man who underwent many rehabilitation activities using ReMoVES for 49 days, completing a total of 41 STS sessions. Group B consists of twenty frail elderly individuals with diverse clinical profiles, including various pathologies and differing levels of disease severity. Some reside in nursing homes in Genoa. A number of them participated in the Rotary Club District 2032 (Liguria and Lower Piedmont, Italy) service project, "Digital Medicine for Prevention and Care." Several members of the group have experienced fractures and orthopedic issues, particularly in the lower limbs. The group comprised eight females and twelve males, with an average age of 82.5 ± 3.4 years. These individuals also performed the STS multiple times, for a total of 34 sessions. In conclusion, 125 sessions are analyzed in total in the present work.

B. Virtual Joint

Since many rehabilitation activities, such as the STS exercise, primarily involve vertical movement of the body, we propose introducing a virtual point representing a sort of center of gravity for a synthetic and better analysis of the signals. Taking the right and left hips (i.e., RHP and LHP joints, respectively) together with the spine middle joint (i.e., SPM joint), the so-called "center of mass (CoM)" virtual joint is defined as:

$$\mathbf{v}_{CoM} = \begin{bmatrix} \frac{x_{SPM}+x_{RHP}+x_{LHP}}{3} \\ \frac{y_{SPM}+y_{RHP}+y_{LHP}}{3} \\ \frac{z_{SPM}+z_{RHP}+z_{LHP}}{3} \end{bmatrix} \tag{8}$$

Owing to this definition, two advantages are gained: improved evaluation of the actual center of gravity and a notable reduction in error, due to the independence of the original joints. We find that the error affecting the CoM $i-$axis is in fact:

$$\epsilon^2_{i_{CoM}} = \frac{\epsilon^2_{i_{SPM}} + \epsilon^2_{i_{RHP}} + \epsilon^2_{i_{LHP}}}{9} \tag{9}$$

which reduces the original error.

C. Kinect Noise Models

The additive and multiplicative noise of the Kinect have been analyzed empirically, and the following conclusions can be drawn, which are also based on the SoA [34]. Referring to Equation 2, the additive term $\{\mathbf{n}_j(t)\} = \{n_{ij}(t)\}$ represents the jitter, which is recognized as an independent disturbance added to the original signal. This noise is characterized by a white spectrum, as anticipated above. As a consequence, it is a zero-mean 3-component vector signal for each joint j. The covariance matrix defined as:

$$\Sigma_{n_{jk}}(i,l) = E\{\mathbf{n}_j(t)\cdot\mathbf{n}_k(t)^T\} = E\{n_{ij}(t)\cdot n_{lk}(t)\}\ \forall i,l,j,k \tag{10}$$

is null for each $j \neq k$, and is diagonal, but its elements are not necessarily equal as in the case of independent, identically distributed (i.i.d.) noise. In fact, ∀ joints j and ∀ axis i, l, it does not necessarily hold that $\Sigma_{n_{jj}}(i,i) = \Sigma_{n_{jj}}(l,l)$. In addition, $\Sigma_{n_{jj}}(i,i)$ is the variance and noise signal mean power at one time, $\sigma^2_{n_{ij}}$. On the basis of a statistical analysis of the Kinect signal, as is also reported in the literature, a loss of the signal at the sensor is experienced, which randomly occurs at some independent instant times and can be modeled as multiplicative noise. A good statistical model refers to an unmodulated Poisson process made of ideal Dirac pulses occurring at Poisson random time instants t_m's with a rate governed by the Poisson density λ. This process is provided directly as input to a filter whose response is a rectangular pulse of random length q. The output $\{w(t)\}$ is then a temporal sequence of randomly placed rectangles of unitary amplitude and random length. The loss of signal samples occurring at the generic axis i of the joint j is then the multiplicative model: $\{m_{ij}(t)\} = \{1 - w(t)\}$.

In conclusion, this noise clearly becomes a binary stationary process, with $P_0 = P\{m(t) = 0\} = P\{w(t) = 1\}$.

D. Signal Standard Error

From Equation 1 and 2, it follows that the signal is

$$\{v_{ij}(t)\} = \begin{cases} \{v_{Bij}(t) + n_{ij}(t)\} & \{m_{ij}(t)\} = 1 \\ 0 & \{m_{ij}(t)\} = 0 \end{cases} \tag{11}$$

As a consequence, the mean square error (MSE) due to only Gaussian noise is $\epsilon^2_{ij}|\{m_{ij}(t) = 1\} = E\{n_{ij}(t)^2\} = \sigma^2_{n_{ij}}$ and the MSE in the case of only multiplicative noise is $\epsilon^2_{ij}|\{m_{ij}(t) = 0\} = E\{v_{Bij}(t)^2\} = S^2_{ij}$.

Finally, applying the total probability theorem, the total error is $\epsilon^2_{ij} = \epsilon^2_{TOT} = P_0 \cdot S^2_{ij} + P_1 \cdot \sigma^2_{n_{ij}}$.

E. Signal Analysis

Before feature extraction and analysis, a preprocessing denoising step is executed, followed by a temporal signal segmentation step, as described in the next subsections.

1) Noise Filtering: As anticipated in section III-B, the issue of missing samples, which causes gaps in the skeletal signals and can be modeled as multiplicative noise, must be addressed before the actual smoothing phase. Therefore, a two-stage preprocessing approach for the joint signals is proposed. The approach first reduces multiplicative noise and then mitigates the jitter effects, modeled as additive noise. As described in subsection IV-C, when samples are missed, runs of zeros appear in the joint signal. These zero-runs cannot be eliminated without risking a loss of synchronization between the signals and the user's actions. Consequently, the preliminary action in the preprocessing step is the detection of these zero-runs, taking into account that their length corresponds to the previously described random variable, q whose expected value ranges from 5 to 8. A cubic spline curve is then fitted using the available non-zero samples to recover the missing portions of the signal, ensuring smooth and continuous transitions in the data. After that, in accordance with the considerations discussed in Section III-B, a low-pass filter is applied to smooth the signal and reduce the jitter effects. A second-order Butterworth filter is employed as a good compromise for reducing noise. Higher-order filters have steeper transitions in the frequency response but may introduce phase shifts, which could affect the synchronization between the various joint signal components. Given the specific movements analyzed, the cutoff frequency was set at 2 Hz. After the preprocessing phase, taking into account multiplicative noise removal and disregarding the noise reduction by the linear filter, one can estimate that the error is reduced to $\epsilon^2_{ijF} = P_1 \cdot \sigma^2_{n_{ij}}$. The expected improvement is reflected in the error reduction factor:

$$\frac{\epsilon^2_{ijF}}{\epsilon^2_{ij}} = \frac{P_1}{1 + P_0 \cdot (SNR - 1)} \tag{12}$$

where the SNR is the signal-to-noise ratio defined as $SNR = \frac{S^2_{ij}}{\sigma^2_{n_{ij}}}$. Notably, the error decreased more consistently in the case of a large SNR and large P_0 values.

2) Signal Segmentation: Rehabilitation movements are typically repetitive, leading to acquired signals that show a certain level of pseudoperiodicity. Movements can be composed of two or more elementary actions or phases, including flexion–extension, adduction–abduction, or transitions between standing and sitting. As a consequence, to ensure a reliable analysis, conducting a signal segmentation phase is crucial. This phase concentrates on stationary segments of the signal that correspond to a homogeneous state, referred to as Γ. The segmentation process results in breaking down complex movements into their elementary actions and must be specifically tailored for each exergame, depending on the required movements and the relevant joints involved. The execution of a generic movement M_α gives rise to a realization of the stocastic process associated with the given experimental trial. The multidimensional random process $\{M_{\alpha,j}(t)\}$ is thus generated. As seen previously, through ReMoVES, it is possible to recover the current game parameters even if the training sessions are started asynchronously at the most convenient time for the patient. Clinicians can easily retrieve the start time of a training session along with events that depend on the patient's performance. This information is also composed of multidimensional signals that are acquired synchronously with the signals described above. Since the different trigger events refer to different movements that the patient performs following the indications of the system, this information has the task of identifying the specific instant in which the specific movement should start (for example, α, β, or others). Guided by the pseudorepetitiveness of the movements, the segmentation process is then devoted to detecting the trigger events observed, which are closely related to the exergame under analysis. In this way, the anticipated movements could be of a different kind, namely, $\{W_\alpha(t)\}$ and $\{W_\beta(t)\}$, respectively, representing the expected signal that should be observed in the correct execution of the movement. As seen above, the completion of sit-to-stand and stand-to-sit movements is verified by analyzing the vertical displacement of the user body through the analysis of the virtual CoM joint defined in Equation 8. The identification of the activation times corresponding to reaching the standing and sitting positions is then based on the vertical speed of the CoM, computed with the discrete incremental ratio. When the person is standing or sitting, the vertical speed is approximately zero; when the person executes the sit-to-stand transition, the speed increases, whereas when the person sits down, the speed is negative. The local maxima and minima of the center of mass are then identified as the zero-crossings of the CoM speed, as shown in Figure 1. Zero crossings with positive derivatives correspond to the start time of the UP transition, and zero crossings with negative derivatives are at the start of the DOWN transition, as shown with stars. All the available signals can subsequently be automatically split into a UP phase (i.e., M_{UP}) and a DOWN phase (i.e., M_{DOWN}) so that features can be computed separately with respect to these basic movements. In such a way, the number of repetitions each subject performs during a game session is automatically extracted (NSU feature).

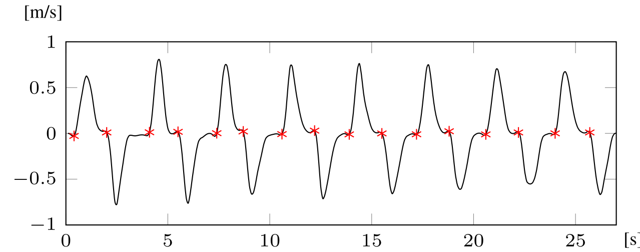

Fig. 1. Zero-crossing of the vertical CoM speed.

F. Clinically Relevant Features

According to the literature, as described in [13], various parameters related to UP and DOWN transition performance have been identified as potentially meaningful in explaining certain disabilities and impairments when evaluating chair-related tests in real clinical settings. The transition duration is the most commonly used metric, along with the vertical linear velocity. High-fall-risk individuals and frail subjects exhibit longer and more variable transition durations. Given that the duration of the STS exergame in ReMoVES is fixed, we evaluate the number of repetitions completed in 30 seconds. We analyze the time required to complete the UP and DOWN transitions and, consequently, the velocity, which is inversely proportional. Other kinematic parameters have also been highlighted in the literature as potential indicators of movement impairments or disabilities. In particular, the trunk tilt range has been shown to distinguish between various levels of frailty. To this end, we have thoroughly analyzed trunk flexion and mediolateral tilt not only in the sitting and standing positions but also during movement. Additional angles related to potential compensatory movements are also evaluated. Reference [9] provided a comprehensive description of the sagittal trunk, hip, knee, and ankle joint ROM, as well as the AP and ML COM displacement during STS exercises, comparing elderly individuals at high risk of falling to others. During the sit-to-stand transition, older fallers presented increased trunk joint ROM in the flexion and extension phases, reduced hip joint ROM in the extension phase, and a greater ML COM shift. During the stand-to-sit transition, they exhibited increased trunk ROM in the flexion phase, reduced hip and knee ROM in the flexion phase, and, again, a greater ML COM displacement. Fallers also took longer to perform stand-to-sit movements, demonstrating more body sway and different joint ROM distributions than nonfallers did. Overall, fallers had more difficulty with stand-to-sit transitions and struggled to control ML motion. On the basis of this rationale, in addition to analyzing the trunk flexion angle in the sitting and standing positions, the angles and range of motion during the UP and DOWN transitions are also considered, along with the ML tilt throughout the entire movement. Shoulder angles also indicate deviations from a correct vertical and straight posture during exercise, representing a clinically relevant parameter and suggesting potential compensatory movements. Knee angles are used to evaluate correct posture in both sitting and standing positions, as well as their correlation during transitions, to assess symmetrical movement. Finally, to segment the complex transition movements into elementary components, the trunk angle is analyzed throughout the movements, and similarities between repeated movements are measured using the dynamic time warping (DTW) feature [35].

G. Feature Extraction

Referring to the trunk flexion angle defined in Equation 3 and on the basis of the considerations that lead to Equation 7, we observe that when the user is in a vertical position, $\theta_{FA-trunk}$ is approximately $\pi/2$; therefore, the measure applies the arccotangent in place of the arctangent. Taking into account that Δz is null in the vertical position and $|\Delta z| = |\Delta y|$ at a maximum theoretical flexion of $\pi/4$, the square error is

$$\epsilon^2_{\theta FA-trunk} = \begin{cases} \dfrac{\epsilon^2_{\Delta z}}{\Delta y^2} & \text{at vertical position} \\ \dfrac{\epsilon^2_{\Delta z} + \epsilon^2_{\Delta y}}{\Delta y^2} & \text{at theoretical maximum flexion} \end{cases} \tag{13}$$

Other angles depicting the deviation with respect to the required vertical or horizontal directions are also computed in terms of the absolute angles. For example, compensatory movements are observed through lateral body angles or shoulder rotations. The angle of the trunk in the frontal plane with respect to the vertical axis takes into account lateral deflection during movement and is computed as:

$$\theta_{LAT-trunk} = \theta\left(\frac{x_{SPS} - x_{SPB}}{y_{SPS} - y_{SPB}}\right) \tag{14}$$

The shoulder angle in the frontal plane with respect to the horizontal line takes into account the inclination of the line joining the right and left shoulders (RS and LS):

$$\theta_{H-shoulder} = \theta\left(\frac{y_{RS} - y_{LS}}{x_{RS} - x_{LS}}\right) \tag{15}$$

The shoulder twist angle (forward and backward) takes into account the AP displacement of the shoulders:

$$\theta_{TW-shoulder} = \theta\left(\frac{z_{LS} - z_{RS}}{x_{RS} - x_{LS}}\right) \tag{16}$$

1) Instant Features: The main clinically relevant features at rest positions are the angles, specifically the knee angles and the torso (or trunk) angles. In the standing position, the knee angles $\psi_{1R} = \{\theta_{RK}|R_{stand}\}$, $\psi_{1L} = \{\theta_{LK}|R_{stand}\}$ and upper body angle $\psi_3 = \{\theta_{FA-trunk}|R_{stand}\}$ are defined and should correspond in the ideal case to the maximum extension. At the sitting position, the knee angles, i.e., $\psi_{2R} = \{\theta_{RK}|R_{sit}\}$, $\psi_{2L} = \{\theta_{LK}|R_{sit}\}$ should be at a minimum value, thus reflecting good flexion. The sitting trunk angle ψ_4 is expected to be large. The feature ψ_0 is the stand-up repetition count in the given exergame session (NSU). Owing to the N repetitions found in a session related to N independent measurements at each rest position, the standard error associated with these features is in fact reduced by a factor of $1/\sqrt(N)$.

2) Dynamic Analysis: Dynamic features are related to complex or single movements M_α from where one can extract the following:

1) Joint displacement along the axes,
2) Angle variations,
3) ROM in terms of the angle variation,
4) Trajectory of single joints,
5) Movement patterns,
6) Phase durations.

The analysis of a dynamic feature becomes essential for evaluating the quality of an elementary movement M_α, such as flexion, abduction, etc., and can only be performed using the above-described temporal segmentation of the signals. Referring to the segmented UP and DOWN transitions described in Section IV-E2, the maximum AP displacement of the trunk, or equivalently, the minimum trunk angles during the ascending and descending phases, explains the flexion required for correct execution of the whole movement and is of outmost clinical relevance. Then, the flexion trunk ROM UP phase and the flexion trunk angle UP phase are equal to $\psi_5 = max\{-\Delta z|M_{UP}\}$ and $\psi_{5_\theta} = min\{\theta_{FA-trunk}|M_{UP}\}$, and similarly for the DOWN phase, they are represented by ψ_6 and ψ_{6_θ}. Features related to undesired compensatory movements are easily evaluated through the ROM computed both as displacement and angle variations. The feature ψ_7 describes the mediolateral inclination of the trunk (ML trunk ROM), which, in principle, must be avoided during both standing and sitting movements to maintain the vertical position: $\psi_7 = ROM\{\theta_{LAT-trunk}\} = max\{max\{\theta_{LAT-trunk}|M_{UP}\}, \quad max\{\theta_{LAT-trunk}|M_{DOWN}\}\}$. The features ψ_8 and ψ_9 represent the rotation of the shoulder axis compared with the horizontal required position and the shoulder rotation in the axial plane, respectively: $\psi_8 = ROM(\theta_{H-shoulder}), \psi_9 = ROM(\theta_{TW-shoulder})$. The average ascending time and descending time give rise to the features ψ_{10} and ψ_{11}, respectively. The feature ψ_{14} is the correlation coefficient of the right and left knee signals.

3) Dynamic Time Warping: As explained above, a cost is calculated that quantifies how the repeated movements are dissimilar, with a low DTW value suggesting greater similarity. Owing to its clinical relevance, the frontal trunk flexion pattern is considered for the UP and DOWN phases: $W_{UPtrunk}(t) = \{\theta_{FA-trunk}(t)|M_{UP}\}$ and $W_{DOWNtrunk}(t) = \{\theta_{FA-trunk}(t)|M_{DOWN}\}$. The features ψ_{12} and ψ_{13} are obtained as the average of the DTW distance measured during a session.

H. Feature Reliability and Meaningfulness

The control group described in Section IV-A was taken into account to assess the reliability of the proposed approach. The following steps have been planned and executed:

TABLE I
FEATURE VALUES FOR CONTROL AND PATIENT GROUPS

Feature		**Mean ± STD**		
	Instant Features	Control	Stroke	Elderly
ψ_0	Repetitions (NSU)	8.42 ± 2.18	8.71 ± 2.79	4.03 ± 1.96
ψ_{1R}	Standing Right Knee Angle [°]	173.30 ± 5.29	170.35 ± 8.08	173.14 ± 6.91
ψ_{1L}	Standing Left Knee Angle [°]	172.53 ± 5.68	169.2 ± 8.39	174.67 ± 5.64
ψ_{2R}	Sitting Right Knee Angle [°]	86.61 ± 10.71	85.94 ± 7.23	84.75 ± 15.37
ψ_{2L}	Sitting Left Knee Angle [°]	85.89 ± 9.96	84.96 ± 7.47	85.80 ± 17.54
ψ_3	Standing Trunk Angle [°]	95.88 ± 7.21	90.28 ± 8.67	86.84 ± 6.79
ψ_4	Sitting Trunk Angle [°]	91.39 ± 10.01	88.83 ± 9.97	86.57 ± 11.75
	Dynamic Features			
ψ_5	Flexion Trunk ROM UP-phase [cm]	16.93 ± 4.98	12.47 ± 3.88	23.76 ± 7.41
ψ_{5_θ}	Flexion Trunk Angle UP-phase [°]	73.96 ± 5.84	76.94 ± 7.87	64.84 ± 7.73
ψ_6	Flexion Trunk ROM DOWN-phase [cm]	20.65 ± 6.93	15.50 ± 6.05	32.72 ± 10.12
ψ_{6_θ}	Flexion Trunk Angle DOWN-phase [°]	74.36 ± 6.99	77.27 ± 8.20	63.83 ± 10.32
ψ_7	ML Trunk ROM [cm]	5.35 ± 1.42	11.04 ± 3.46	14.02 ± 22.51
ψ_{7_θ}	ML Trunk Angle [°]	5.58 ± 1.97	11.1 ± 6.94	13.4 ± 6.12
ψ_8	H Shoulder Angle [°]	7.38 ± 3.19	10.7 ± 4.67	13.5 ± 15.33
ψ_9	TW Shoulder Angle [°]	7.82 ± 2.56	15.4 ± 5.19	20 ± 17.90
ψ_{10}	Ascending Time [sec]	1.51 ± 0.39	1.51 ± 0.59	4 ± 2.49
ψ_{11}	Descending Time [sec]	1.73 ± 0.46	1.69 ± 0.66	4.09 ± 2.28
ψ_{12}	Ascending DTW	1.14 ± 0.77	1.69 ± 0.96	2.57 ± 1.52
ψ_{13}	Descending DTW	1.97 ± 0.89	2.36 ± 1.04	4.55 ± 2.20
ψ_{14}	Knee Correlation	0.9988 ± 0.0014	0.9732 ± 0.0503	0.9960 ± 0.0032

- homogeneity of the control group for each feature (interrater reliability);
- homogeneity of feature values extracted at the repetitions within a single game session (test–retest reliability, intrarater reliability);
- correlation analysis for symmetrical joints in symmetrical movements;
- similarity of movement patterns among repetitions of a single session (test–retest);
- similarity of movements for different sessions.

On the basis of the control group and patients, each proposed feature is then analyzed with the aim of understanding its significance and potential applicability. Finally, in Section VI-B, a discussion of feature error is given.

V. RESULTS

The results presented refer to the above-described groups of healthy subjects (26 sessions), stroke patients (65 sessions), and elderly individuals (34 sessions), totaling 125 sessions. The features extracted and analyzed for the considered groups are the ones described above, from ψ_0 to ψ_{14} and reported in Table I. Each feature is measured from the joint signals as described in Subsection IV-G. Except for the ROM and minimum values, the values extracted from several repetitions are averaged to reduce the error. The first subsection (i.e., V-A) outlines the approach used to demonstrate the robustness and meaningfulness of the automatically extracted features over extended periods, which may be useful for monitoring patient evolution. Subsection V-B provides a description of the features related to the control group of healthy subjects, serving as a reference for the correct execution of the STS exercise. Subsection V-C, while detailing the features acquired during the stroke and elderly patient sessions, initiates a discussion of robustness across groups, which will be further elaborated upon in the Discussion section.

A. Reliability and Meaningfulness

Given the small sample size, data normality was assessed via the Shapiro–Wilk test [36]. For all the parameters computed from the control group sessions, the results of the test confirmed the acceptance of the normality condition ($p > 0.01$). In contrast, the test rejected the normality hypothesis for most of the parameters, with $p < 0.001$, for both groups of stroke patients and elderly individuals, with a few exceptions. The flexion knee angles were normal for both groups, the extension and flexion trunk angles were normal for the elderly group, and the shoulder twist angle was normal for the stroke group. In healthy subjects, the correlation between the instantaneous angles of the right and left knees was taken as an index of feature consistency. The correlation coefficient (Pearson) evaluated for the control group was found to be equal to $\rho = 0.9988$, confirming the strong robustness of the feature, as one might have expected from the above considerations of the measurement error. Surprisingly, when the same correlation index was computed for the other groups, we found $\rho = 0.9732$ and $\rho = 0.9960$ for the stroke patients and elderly patients, respectively. One explanation is that the patients considered, despite their age or the outcome of neurological events, have a high degree of autonomy and are in good physical condition. Since normality is not always true, with the purpose of discussing significant differences between the groups, the nonparametric Mann–Whitney U-test for independent samples was applied. For all tests, the null hypothesis H_0 assumes that the patient's value of a given parameter does not differ from that of the control group. Depending on the parameter considered, three different alternative hypotheses H_1 were defined, referring to one-sided or two-sided tests. With respect to intrarater reliability, the similarity of movement patterns among repetitions of a single session was evaluated for the control group through the variance of the DTW values obtained by single repetitions within a session.

B. Control Group Characterization

1) Feature Analysis: After the segmentation step described in Section IV-E2, for each game session, the values of the instant features of interest (knee angles and trunk angles, ψ_1, ψ_2, ψ_3, and ψ_4) were calculated for each single repetition, and then the values were averaged over the number of repetitions (i.e., ψ_0 = NSU). The same holds for duration times ψ_{10} and ψ_{11}. However, in the movements of the trunk, the ROM and minimum values are computed for ψ_5 and ψ_6. The same holds for the trunk and shoulder movements, which are considered to account for compensatory motions that should remain at low values for the entire duration of the session. The values extracted are summarized in Table I, together with the computed standard deviations (STDs). As reported in previous works, the number of repetitions in the 30-s STS [24] mainly depends on age for healthy subjects, in addition to physical prowess, reaction times, etc. Figure 2 shows the instantaneous knee angle for both the right (blue line) and left (red line) limbs during one session performed by one of the healthy subjects. The standing position requires the legs to be fully extended (at the peaks of the angle signal), and in fact, they reach an average of about 173° at the point of the standing position, indicated by the green cross, as detected through the previously described CoM analysis. When the person is seated, this angle is varying between 80° and 90° (indicated by the orange crosses during sitting times). We can observe that healthy

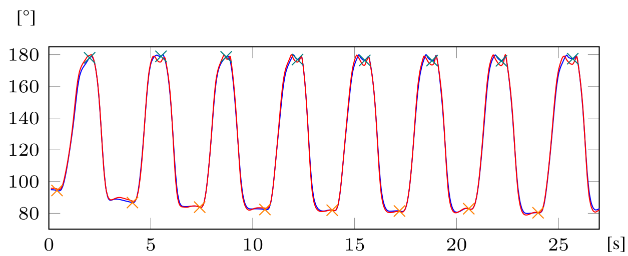

Fig. 2. Instantaneous knee angle during a healthy subject session. The red line represents the left limb, and the blue line represents the right limb. The green crosses represent standing positions, and the orange crosses represent sitting positions.

subjects present a near-neutral trunk position in the frontal plane, with a small lateral tilt, which is measured on the order of 5°. The subject is also required to preserve shoulders in the frontal plane, i.e., without twisting or inclinations. Indeed, in the control group, the degree of shoulder displacement in the sagittal and transverse planes was very small, indicating good control of the standing and sitting movements, which were performed without any compensation. In the healthy group, the pattern of the antero-posterior trunk angle was fairly regular. When the subject is in the seated position.

The trunk angle is approximately 90°; when the subject is in the standing position, the trunk angle is observed to be slightly larger, as reported for feature ψ_3 in Table I. During the ascent and descent phases, the flexion of the trunk causes these angles to decrease to approximately $70° - 80°$. In both the ascending and descending phases, the movements show strong repetitiveness, both in relative measures and absolute values. The DTW mean values for the control group are reported in Table I and are always smaller than 4.5 for both the UP and DOWN phases. As an important outcome of this analysis, we can define the empirical description of the correct execution of the movement required by the exergame. The feature values computed as the average of the control group are the reference model to be used for comparison with other subjects and patients. It is possible to monitor the activity of patients and evaluate whether a person is correctly performing the required movement, their deviations, etc. Such a possible application is described in the following paragraph with respect to Groups A and B. As a general comment, one can observe from Table I that the standard error in the angle features of the control group is always smaller than 10.8, which accounts for both subjects variability and measurement error. A discussion about the RMSE of such features is given in Section VI.

C. Patient Analysis

The Mann–Whitney U-test is applied for each feature. The null hypothesis H_0 is that the patient's value does not differ from that of the control group, where the mean value is taken as a reference: $X_{patient} = \mu_{control}$. The test is repeated for each feature for Group A (stroke patients) and Group B (elderly individuals).

1) Number of Repetitions: As described in Subsection IV-F, we expect both patient groups to exhibit a smaller and more variable number of transitions than the control group. The observed mean repetition values (i.e., NSU, ψ_0) are 8.42 for healthy individuals, 8.71 for stroke patients, and 4.03 for elderly individuals. A statistical test revealed a significant difference between the control group and Group B of frail elderly people, with $p < 0.001$, indicating a greater risk of falls, as suggested in [11]. In contrast, the value for the stroke group did not differ significantly from that of the healthy subjects (with a p value of 0.553) and was even greater. However, when evaluating the expected larger variation in this feature among the groups, as measured by the coefficient of variation (i.e., standard deviation divided by the mean), we find significant differences, with 25%, 32%, and 48% for healthy individuals, stroke patients, and elderly individuals, respectively.

2) Ascending and Descending Times: Considering the time to stand up and the time to sit (ψ_{10} and ψ_{11}), stroke patients behave like healthy subjects. However, for elderly people, there appears to be a significant difference, with $p < 0.001$. Table I also shows that the standing times are generally shorter than the sitting times in all the groups.

3) Knee Angles: For the majority of patients, the flexion and extension angles of the legs are comparable with respect to those of healthy subjects. The minimum p-value is 0.059 for stroke patients, who tend to have less extended legs when standing, as shown in Table I (the ψ_1 features).

4) Antero-Posterior Trunk Motion: From Table I, the features ψ_3 and ψ_4 show how patients tend to show similar trunk extension angles when standing and sitting, unlike healthy people, who extend their torso more when standing. Compared with the control group, when the trunk angle during the seated rest position was considered, there were no significant differences between the two groups. In contrast, the trunk flexion angle in the standing position was partially significant in Group A ($p - value = 0.003$) and more significant ($p - value < 0.001$) in Group B, as explained by the common posture of elderly and stroke patients reported in the literature.

5) Dynamic Features: When analyzing dynamic features, the movements during the ascending and descending transitions of a single session are examined for the patients, as previously conducted for the control group. As shown in Table I by the features ψ_5 and ψ_6, elderly patients tend to flex the trunk more during the ascent and descent phases, as described in Subsection IV-F, to help movement and compensate for lower limb weakness (the absolute trunk angle with respect to the horizontal plane is approximately 64 degrees). Our stroke patients generally follow a pattern quite similar to that of healthy subjects; in contrast, frail elderly people usually perform fewer repetitions, and their movements deviate significantly from the standard. In Section III of the Supplementary File, figures illustrating examples of trunk flexion angles during ascending and descending transitions are presented.

6) Shoulder and Lateral Trunk Movements: While healthy subjects present a near-neutral trunk position in the frontal plane, poststroke subjects show a trunk tilt toward the less affected side during STS when rising from a chair using spontaneous foot positions [37]. This trunk shift was also observed before leaving the seat in [38] and [39] and was estimated to be 12.1° ±6.1, whereas it was 2.4° in healthy subjects [40]. As described in Section II-A, a lateral tilt

TABLE II
MANN–WHITNEY $p-$ *values* FOR THE ELDERLY AND STROKE PATIENT GROUPS AND COHEN'S d EFFECT SIZE

Alternative Hypothesis	Feature	Mann-Whitney p-value		Cohen's d	
		Stroke	Elderly	Stroke	Elderly
$X_{patient} < \mu_{control}$	ψ_{1R}	0.083	0.636	0.39	0.02
	ψ_{1L}	0.059	0.955	0.43	0.37
$X_{patient} \neq \mu_{control}$	ψ_0	0.553	<0.001	0.11	2.13
	ψ_{2R}	0.645	0.394	0.08	0.13
	ψ_{2L}	0.689	0.819	0.11	0.005
	ψ_3	0.003	<0.001	0.67	1.29
	ψ_4	0.392	0.116	0.25	0.43
	ψ_5	0.013	<0.001	0.40	1.30
	ψ_6	0.045	<0.001	0.36	1.16
	ψ_{10}	0.195	<0.001	0.001	1.31
	ψ_{11}	0.370	<0.001	0.06	1.35
$X_{patient} > \mu_{control}$	ψ_8	<0.001	0.001	0.77	0.52
	ψ_9	<0.001	<0.001	1.65	0.89
	ψ_7	<0.001	<0.001	0.92	1.62

of the trunk is usually observed in poststroke subjects. The value of $\psi_{7\theta}$ observed here is approximately 11°, which is much larger than that observed in the control group (i.e., 5.58°). Even greater trunk displacement was observed for older adults (i.e., 13.4°). Table I shows that the patients showed significant differences in shoulder alignment throughout the entire duration of the exercise. Finally, we can conclude that the patients' impairments, due to both poststroke outcomes and frailty or orthopedic problems, create difficulties that are compensated with an asymmetric movement, resulting in twisting and rotation of the shoulders.

VI. DISCUSSION

The current feasibility study is small scale and focuses primarily on assessing the practicality and potential challenges of using ReMoVES for future clinical trials rather than testing the effectiveness of the intervention itself, an aspect that has already been demonstrated by previous research. The study does not aim to evaluate therapeutic outcomes but instead monitors the execution of exercises and movement patterns, particularly aiming to predict remotely whether each execution is good or poor. In the following subsections, effect size and error evaluation are discussed in relation to the proposed method and features, and their adequacy is assessed. A discussion on sample size is reported in the Supplementary File, section IV. These concepts are crucial for ensuring that the study's findings are robust and not compromised by an inadequate sample size or poor error management. The final subsections explore aspects of practical applicability and relevance, as well as future perspectives, with a particular focus on non-clinical settings.

A. Effect Size

The effect size was assessed by measuring Cohen's d as reported in Table II and taking into account the related meaning as established by Sawilowsky [41] (i.e., very small for d close to 0.01; small for d close to 0.2; medium for d close to 0.5; large for d near 0.8; very large for d close to 1.2; and huge for d equal to or greater than 2). With respect to the elderly group, one can observe a huge effect for the number of repetitions, a large effect for shoulder twist, and a very large effect for both the UP and DOWN times and all trunk angle features (except for the sitting trunk angle, which is not significant with a p value of 0.116), as one might expect on the basis of the previous discussion in Subsection IV-F. A medium-level effect is associated with the horizontal shoulder angle. With respect to the stroke group, a very large effect is observed for shoulder twist, a large effect is observed for the ML trunk displacement angle, and a medium effect is observed for the standing trunk angle and horizontal shoulder angle.

B. Error Evaluation

As described in Section IV-E1, nonlinear filtering allows a reduction in the error at each joint tracking, as defined in Equation 12. Considering that the temporal density of the Poisson points in the time unit is equal to λ, by construction, we find that $P_0 = \lambda \cdot E\{q\} = \lambda \cdot \overline{q}$. The reasonable values for λ are on the order of 1 in a thousand; for the random variable q, the typical expected value ranges from 5 to 8. As a consequence, the following discussion takes into account P_0 values in the range of 0.001–0.01, with special cases of P_0 being equal to 0.001, 0.005, and 0.008. With respect to the CoM error, some considerations can be made on the basis of previous works in the literature, which give error estimates for joint measurements. According to Table II of [24], if we take the difference between Kinect and Qualisys as a rough, conservative estimate of the Kinect error, the RMSE values reported for the hips and spine middle range from 1.67 to 6.23 cm, which are smaller in the mediolateral axis and larger in the antero-posterior axis (z-axis of the left hip). The mean value is 2.56 for the x-axis, 4.07 for y and 4.67 for z. Our proposal of using the virtual CoM joint reduces the RMSE to 1.53, 2.43, and 2.78, respectively, which is a 40% reduction. A further reduction is obtained with the proposed nonlinear filtering procedure. In the most conservative case of an SNR equal to 10 dB, the CoM joint coordinates are reduced by a further 0.5% in the case of P_0 equal to 0.001, by 2.5% when P_0 is 0.005, and by 3.8% when P_0 is 0.008. The final error is always smaller than 2.8 cm. To discuss the errors associated with angle measurements, let us take the trunk flexion angle as an example. Table II in [24] also reports the errors of the spinal base and shoulder joints. The RMSE of these joints is also reduced. The error associated with Δl is reduced as described in Section III-C1, and the estimated error of the angles is reduced accordingly. As expected, the maximum error occurs for angles close to $\pi/4$ and is strongly reduced in the case of large L values. For a comparison with the state-of-the-art numbers, in the work cited in [24], the error angles for the knee range from 5° to 22° (Table 4), and the average errors in the STS range from 22.7° to 33.1° (Table 5). Here, as seen above (III-C1), owing to the N repetitions found in a session related to N independent measurements, the standard error associated with these features is further reduced by a factor of $1/\sqrt{(N)}$. One finally achieves 50% of the error of the single take in the case of 4 repetitions and 35% of the error in the case of 8 repetitions. The CoM error is always smaller than 1.4 cm, and the angle errors are always smaller than 6 degrees.

C. Non-Clinical Setting

In a home rehabilitation setting, identifying when a patient is executing an exercise incorrectly (for example, repeating a movement too many times without completing the full motion or introducing compensatory movements) can trigger an alert to the therapist team, who can then provide guidance for better execution. In addition to observing the executed activities in comparison to the weekly prescription plan, this observation through the Therapist Client helps the multidisciplinary rehabilitation team adapt the Individual Rehabilitation Plan.
Indeed, rather than expecting immediate improvements in patient functionality, a key contribution for clinical assessment may lie in the long-term monitoring of patient performance. For instance, after discharge from the hospital, it is common to observe a decline in the functionalities regained during the rehabilitation period. Monitoring patient behavior in such cases is crucial for helping maintain the re-acquired functionalities. To enable real-world deployment, the system must address practical challenges related to its usability in non-clinical settings, such as home environments, while also meeting the specifications required for CE marking of Class I medical devices. Firstly, in addition to a printed handbook, short instructional videos are provided to explain the system's usage, gameplay instructions, and overall functionality to the patient and their caregiver. This approach takes into account that the system interface is highly intuitive and user-friendly, requiring no expertise in technology. Home environments often face unreliable internet connections or lack Wi-Fi infrastructure, which can disrupt data transmission. To address this, the system implements a local buffering mechanism that temporarily stores data during connectivity issues and synchronizes it with remote servers once the connection is restored. Additionally, the system allows the patient or caregiver to use a smartphone with a mobile network as a transmission link, with a simple procedure that starts automatically after the game sessions are completed. Remote technical support is offered via video or phone calls, with assistance available within 24 hours.

D. Future Works

As described in [42], the ReMoVES platform is designed to integrate various technologies, including touchscreen interfaces for cognitive exergames and Leap Motion for hand exercises. It was originally developed to incorporate also biophysical sensors, such as the Microsoft Band, which was used at that time. As sensor technology continues to advance rapidly, the open IoT architecture is designed to integrate new sensors at the Application Layer. We have conducted several feasibility studies using the ReMoVES system, focusing on different patient groups. For instance, [32] describes the STORMS project, which targets patients with Multiple Sclerosis, while [43] reports its use with pediatric patients, demonstrating the system's potential for managing these conditions. Additionally, a preliminary study on Systemic Sclerosis patients utilized Leap Motion for hand pathology, showcasing its flexibility in integrating different sensors for specific rehabilitation needs [44]. In terms of telehealth infrastructure, telehealth hubs are designed to centralize and manage remote healthcare data, offering tools for real-time patient monitoring, alerts, and integration with various remote systems. These hubs provide dashboards and interfaces that facilitate monitoring, ensuring continuity of care and effective remote healthcare delivery. In this context, the ReMoVES platform, with its IoT architecture, shows great potential for scalability, particularly in telerehabilitation. The authors aim to extend the system's application to home-based rehabilitation and nursing homes, supporting traditional therapies and remote assessments. The initiative aligns with the PNRR/M6 Health Mission, particularly Component M6C1, which focuses on proximity Community Health Centers [45]. A potential future application of Azure Kinect, briefly outlined in the Supplementary File, may reduce reliance on specialized sensors like Leap Motion for hand and finger movement analysis. Furthermore, even in controlled environments, the sensor can partially mitigate lighting and occlusion issues that may arise. Given the related advantages and drawbacks, integrating additional sensors, such as IMU, EMG, or wearable accelerometers, could enhance the system's robustness and utility. The combination of IMU and Kinect, in particular, could be highly beneficial for achieving drift-free and smooth tracking, capturing finer compensatory movements, and analyzing smaller joint activities. Indeed, practical challenges must be addressed to ensure ease of setup and functionality for home-based use, including issues with sensor positioning, recalibration due to changes in device attachment, and potential time delays during the initialization or calibration phase.

VII. Conclusion

In this study, we explored the application of a markerless telerehabilitation system, ReMoVES, which focuses on the sit-to-stand task, to perform multidimensional signal processing during telerehabilitation sessions, with an emphasis on preprocessing and analysis to extract reliable parameters for remote patient observation. The discussion of the error evaluation highlighted the effectiveness of nonlinear filtering in reducing errors at each signal instant. The proposed method demonstrated significant reductions in the error for various signal-to-noise ratio values, providing valuable insights into noise reduction and its impact on telerehabilitation accuracy. The control group characterization provided a comprehensive analysis of feature distribution, with an empirical description of correct movement execution during telerehabilitation. This reference model enables the monitoring of patient activities, facilitating the identification of deviations and assisting in personalized rehabilitation plans. Even though the current work is a feasibility study, it has also demonstrated the statistical robustness of features within groups. Additionally, a key outcome is the good robustness observed across groups, with the features behaving as expected from a clinical perspective when comparing healthy to frail individuals. In conclusion, this study provides valuable insights into telerehabilitation, presenting a robust methodology that serves as a foundation for the ongoing research in anomaly detection, which is briefly contextualized in the Supplementary File. The established reference model provides a basis for patient monitoring and protocol development, advancing the field toward more personalized and effective telerehabilitation interventions. The ReMoVES platform (featuring the original games described in [46] and the extended set from the STORMS version [32]) has been autonomously installed in the homes of nine patients for a pilot feasibility study on telerehabilitation. The goal was to perform exercises under remote monitoring for at least three weeks. Except for three patients who discontinued follow-up due to clinical reasons, the installation and use of the system were perceived as simple and highly satisfactory. The home setting closely resembles the clinical environment, where a common room or dining area is often utilized, and no significant differences in functionalities or patient performance have been observed. In addition to clinical applications, the growing interest in MMC technology for tracking human movement has driven the development of cost-effective commodity systems. They estimate key body points using statistical methods and AI-based software. 3D systems, such as Microsoft's Kinect, Orbbec's Astra Mini [47], and Intel's RealSense [48], use both a video camera and an infrared depth sensor and enable movement assessments in uncontrolled environments, such as homes or outdoor settings. They also facilitate the collection of larger datasets on patient health or athlete performance, allowing for better tracking of progress after exercise or treatment. Technological advancements in both hardware and software are expected to further expand the reach and effectiveness of rehabilitation tools, making them more accessible and efficient. In conclusion, with the goal of continuing to use simple technology to keep development costs low and enhance potential applicability in real-world environments, the integration of additional sensors, such as inertial sensors, into the system offers significant potential. Further improvements could be achieved through the use of new off-the-shelf products recently introduced to the market. One notable example is the Femto camera, which combines Azure Kinect's ToF technology with Orbbec's video analysis capabilities [49]. Finally, a promising advancement in research focused on anomaly detection, while also addressing privacy concerns and ensuring adaptability to diverse patient populations, can come from machine learning and deep learning techniques.

References

[1] M. Trombini, F. Ferraro, M. Morando, G. Regesta, and S. Dellepiane, "A solution for the remote care of frail elderly individuals via exergames," *Sensors*, vol. 21, no. 8, p. 2719, Apr. 2021.

[2] G. Kurillo, E. Hemingway, M.-L. Cheng, and L. Cheng, "Evaluating the accuracy of the Azure Kinect and Kinect v2," *Sensors*, vol. 22, no. 7, p. 2469, Mar. 2022.

[3] R. W. Bohannon, "Sit-to-Stand test for measuring performance of lower extremity muscles," *Perceptual Motor Skills*, vol. 80, no. 1, pp. 163–166, Feb. 1995.

[4] Y. Oh and S. Yang, "Defining exergames & exergaming," in *Proc. Meaningful Play*, 2010, pp. 21–23.

[5] D. Hebert et al., "Canadian stroke best practice recommendations: Stroke rehabilitation practice guidelines, update 2015," *Int. J. Stroke*, vol. 11, no. 4, pp. 459–484, Jun. 2016.

[6] S. Ranaldi, L. Gizzi, G. Severini, and C. De Marchis, "Optimal identification of muscle synergies from typical sit-to-stand clinical tests," *IEEE Open J. Eng. Med. Biol.*, vol. 4, pp. 31–37, 2023.

[7] C. Ta and R. Jin, "Exploring machine learning techniques for fall risk prediction using the Berg balance scale," in *Proc. IEEE World AI IoT Congr. (AIIoT)*, Jun. 2023, pp. 0273–0277.

[8] R. Ganea, A. Paraschiv-Ionescu, C. Büla, S. Rochat, and K. Aminian, "Multi-parametric evaluation of sit-to-stand and stand-to-sit transitions in elderly people," *Med. Eng. Phys.*, vol. 33, no. 9, pp. 1086–1093, Nov. 2011. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1350453311000968

[9] Y.-T. Lin and H.-J. Lee, "Comparison of the lower extremity kinematics and center of mass variations in sit-to-stand and stand-to-sit movements of older fallers and nonfallers," *Arch. Rehabil. Res. Clin. Transl.*, vol. 4, no. 1, Mar. 2022, Art. no. 100181. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590109522000040

[10] B. K. Shukla, H. Jain, V. Vijay, S. K. Yadav, A. Mathur, and D. J. Hewson, "A comparison of four approaches to evaluate the sit-to-stand movement," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 6, pp. 1317–1324, Jun. 2020.

[11] K. L. Johansen, R. D. Stistrup, C. S. Schjøtt, J. Madsen, and A. Vinther, "Absolute and relative reliability of the timed 'up & go' test and '30second chair-stand' test in hospitalised patients with stroke," *PLoS ONE*, vol. 11, no. 10, Oct. 2016, Art. no. e0165663.

[12] C. Gu, W. Lin, X. He, L. Zhang, and M. Zhang, "IMU-based motion capture system for rehabilitation applications: A systematic review," *Biomimetic Intell. Robot.*, vol. 3, no. 2, Jun. 2023, Art. no. 100097.

[13] N. Millor, P. Lecumberri, M. Gomez, A. Martinez-Ramirez, and M. Izquierdo, "Kinematic parameters to evaluate functional performance of sit-to-stand and stand-to-sit transitions using motion sensor devices: A systematic review," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no. 5, pp. 926–936, Sep. 2014. [Online]. Available: https://api.semanticscholar.org/CorpusID

[14] M. Goffredo, M. Schmid, S. Conforto, M. Carli, A. Neri, and T. D'Alessio, "Markerless human motion analysis in Gauss–Laguerre transform domain: An application to sit-to-stand in young and elderly people," *IEEE Trans. Inf. Technol. Biomed.*, vol. 13, no. 2, pp. 207–216, Mar. 2009.

[15] W. W. T. Lam, Y. M. Tang, and K. N. K. Fong, "A systematic review of the applications of markerless motion capture (MMC) technology for clinical measurement in rehabilitation," *J. NeuroEng. Rehabil.*, vol. 20, no. 1, p. 57, May 2023, doi: 10.1186/s12984-023-01186-9.

[16] B. Lange, C.-Y. Chang, E. Suma, B. Newman, A. S. Rizzo, and M. Bolas, "Development and evaluation of low cost game-based balance rehabilitation tool using the Microsoft Kinect sensor," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2011, pp. 1831–1834.

[17] L. R. A. Dos Santos et al., "The use of nintendo wii in the rehabilitation of poststroke patients: A systematic review," *J. Stroke Cerebrovascular Diseases*, vol. 24, no. 10, pp. 2298–2305, Oct. 2015.

[18] A. Dash, A. Yadav, A. Chauhan, and U. Lahiri, "Kinect-assisted performance-sensitive upper limb exercise platform for post-stroke survivors," *Frontiers Neurosci.*, vol. 13, p. 228, Mar. 2019.

[19] J. Cancela, M. T. Arredondo, and O. Hurtado, "Proposal of a KinectTM-based system for gait assessment and rehabilitation in Parkinson's disease," in *Proc. 36th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Aug. 2014, pp. 4519–4522.

[20] N. Kitsunezaki, E. Adachi, T. Masuda, and J.-I. Mizusawa, "KINECT applications for the physical rehabilitation," in *Proc. IEEE Int. Symp. Med. Meas. Appl. (MeMeA)*, May 2013, pp. 294–299.

[21] J. García, K. F. Navarro, D. Schoene, S. Smith, and Y. Pisan, "Exergames for the elderly: Towards an embedded Kinect-based clinical test of falls risk," in *Proc. HIC*, 2012, pp. 51–57.

[22] M. van Diest, J. Stegenga, H. J. Wörtche, K. Postema, G. J. Verkerke, and C. J. C. Lamoth, "Suitability of Kinect for measuring whole body movement patterns during exergaming," *J. Biomech.*, vol. 47, no. 12, pp. 2925–2932, Sep. 2014.

[23] D. Pagliari and L. Pinto, "Calibration of Kinect for xbox one and comparison between the two generations of Microsoft sensors," *Sensors*, vol. 15, no. 11, pp. 27569–27589, Oct. 2015. [Online]. Available: https://api.semanticscholar.org/CorpusID

[24] A. Napoli, S. Glass, C. Ward, C. Tucker, and I. Obeid, "Performance analysis of a generalized motion capture system using Microsoft Kinect 2.0," *Biomed. Signal Process. Control*, vol. 38, pp. 265–280, Sep. 2017.

[25] K. Otte et al., "Accuracy and reliability of the Kinect version 2 for clinical measurement of motor function," *PLoS ONE*, vol. 11, pp. 1–17, Nov. 2016, doi: 10.1371/JOURNAL.PONE.0166532.

[26] J. F. Gauss, C. Brandin, A. Heberle, and W. Löwe, "Smoothing skeleton avatar visualizations using signal processing technology," *Social Netw. Comput. Sci.*, vol. 2, no. 6, p. 429, Nov. 2021.

[27] Z. Niu, K. Lu, J. Xue, X. Qin, J. Wang, and L. Shao, "From method to application: A review of deep 3D human motion capture," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 11, pp. 11340–11359, Jul. 2024.

[28] D. Winter, *Biomechanics and Motor Control of Human Movement*, 4th ed., Hoboken, NJ, USA: Wiley, 2009.

[29] G. M. Moreira, L. H. F. Giovanini, M. P. R. de Castro, G. N. Nogueira, T. C. Boumer, and E. F. Manffra, "Filtering motion signals from Microsoft Kinect in the context of stroke rehabilitation," *Res. Biomed. Eng.*, vol. 35, pp. 265–270, Dec. 2019.

[30] L. F. Yeung, K. C. Cheng, C. H. Fong, W. C. C. Lee, and K.-Y. Tong, "Evaluation of the Microsoft Kinect as a clinical assessment tool of body sway," *Gait Posture*, vol. 40, no. 4, pp. 532–538, Sep. 2014.

[31] A. Schmitz, M. Ye, R. Shapiro, R. Yang, and B. Noehren, "Accuracy and repeatability of joint angles measured using a single camera markerless motion capture system," *J. Biomech.*, vol. 47, no. 2, pp. 587–591, Jan. 2014.

[32] L. Vestito et al., "STORMS: A pilot feasibility study for occupational TeleRehabilitation in multiple sclerosis," *Sensors*, vol. 24, no. 19, p. 6470, Oct. 2024.

[33] F. Ferraro, G. Iaconi, M. Simonini, and S. Dellepiane, "Signal processing for remote monitoring of home-based rehabilitation support activities," in *Proc. IEEE Int. Conf. E-Health Netw., Appl. Services (HealthCom)*, Oct. 2022, pp. 192–198.

[34] C. V. Nguyen, S. Izadi, and D. Lovell, "Modeling Kinect sensor noise for improved 3D reconstruction and tracking," in *Proc. 2nd Int. Conf. 3D Imag., Modeling, Process., Visualizat. Transmiss.*, Oct. 2012, pp. 524–530.

[35] H. Sakoe and S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-26, no. 1, pp. 43–49, Feb. 1978.

[36] S. W. Cheng and F. A. Spiring, "A test to identify the uniform distribution, with applications to probability plotting and other distributions," *IEEE Trans. Rel.*, vol. R-36, no. 1, pp. 98–105, Apr. 1987.

[37] A. Boukadida, F. Piotte, P. Dehail, and S. Nadeau, "Determinants of sit-to-stand tasks in individuals with hemiparesis post stroke: A review," *Ann. Phys. Rehabil. Med.*, vol. 58, no. 3, pp. 167–172, Jun. 2015.

[38] S. Hesse, M. Schauer, M. Malezic, M. Jahnke, and K. Mauritz, "Quantitative analysis of rising from a chair in healthy and hemiparetic subjects," *Scandin. J. Rehabil. Med.*, vol. 26, no. 3, pp. 161–166, Sep. 1994.

[39] C. Mazza, S. J. Stanhope, A. Taviani, and A. Cappozzo, "Biomechanic modeling of sit-to-stand to upright posture for mobility assessment of persons with chronic stroke," *Arch. Phys. Med. Rehabil.*, vol. 87, no. 5, pp. 635–641, May 2006.

[40] J. Lecours, S. Nadeau, D. Gravel, and L. Teixera-Salmela, "Interactions between foot placement, trunk frontal position, weight-bearing and knee moment asymmetry at seat-off during rising from a chair in healthy controls and persons with hemiparesis," *J. Rehabil. Med.*, vol. 40, no. 3, pp. 200–207, Mar. 2008.

[41] S. S. Sawilowsky, "New effect size rules of thumb," *J. Modern Appl. Stat. Methods*, vol. 8, no. 2, p. 26, Nov. 2009. [Online]. Available: https://api.semanticscholar.org/CorpusID:117432360

[42] M. Morando, S. Ponte, E. Ferrara, and S. Dellepiane, "Definition of motion and biophysical indicators for home-based rehabilitation through serious games," *Information*, vol. 9, no. 5, p. 105, May 2018.

[43] H. Hotiet, C. Baffigo, R. Truffelli, M. Simonini, and S. Dellepiane, "Virtual reality and markerless motion capture in children rehabilitation," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Montreal, QC, Canada, Jun. 2025.

[44] F. Ferraro, M. Trombini, M. Morando, M. Doveri, G. Bianchi, and S. Dellepiane, "Exergames for systemic sclerosis rehabilitation: A pilot study," in *Advances in Computer Vision and Computational Biology* (Transactions on Computational Science and Computational Intelligence), H. Arabnia, L. Deligiannidis, H. Shouno, F. Tinetti, and Q. Tran, Eds., Cham, Switzerland: Springer, 2021, doi: 10.1007/978-3-030-71051-4_22.

[45] Ministero della Salute. *PNRR-Salute: Investimenti E Progetti*. Accessed: Feb. 6, 2025. [Online]. Available: https://www.pnrr.salute.gov.it/portale/pnrrsalute/dettaglioContenutiPNRRSalute.jsp?lingua=italiano&id=5801&area=PNRR-Salute&menu=investimenti

[46] M. Trombini et al., "A study protocol for occupational rehabilitation in multiple sclerosis," *Sensors*, vol. 21, no. 24, p. 8436, Dec. 2021.

[47] Orbbec. (2024). *Astra Mini*. Accessed: May 15, 2022. [Online]. Available: https://orbbec3d.com/astra-mini

[48] Intel. (2024). *Intel Realsense*. Accessed: May 15, 2022. [Online]. Available: https://realsense.intel.com

[49] Orbbec. *Microsoft Collaboration for Innovative Sensor Technology*. Accessed: Oct. 21, 2024. [Online]. Available: https://www.orbbec.com/microsoft-collaboration/